

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. ReportExample2.
AUTHOR. Michael Coughlan.
```

Click to view the program
commentary

```
ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
```

```
SELECT SalesFile ASSIGN TO "GBSALL"
    ORGANIZATION IS LINE SEQUENTIAL.
SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".
```

```
DATA DIVISION.
FILE SECTION.
```

```
FD SalesFile.
01 SalesRecord.
   88 EndOfFile VALUE HIGH-VALUES.
   02 CityCode PIC 9.
   02 SalesPersonNum PIC 9.
   02 ValueOfSale PIC 9(4)V99.
```

```
FD PrintFile
REPORT IS SalesReport.
```

```
WORKING-STORAGE SECTION.
```

```
01 NameTable.
   02 TableValues.
      03 FILLER PIC X(18) VALUE "Dublin Belfast ".
      03 FILLER PIC X(18) VALUE "Cork Galway ".
      03 FILLER PIC X(18) VALUE "Sligo Waterford".
      03 FILLER PIC X(9) VALUE "Limerick".
   02 FILLER REDEFINES TableValues.
      03 CityName PIC X(9) OCCURS 7 TIMES.
```

```
REPORT SECTION.
```

```
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

```
$ SET SOURCEFORMAT"FREE"
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```
ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
```

```
SELECT SalesFile ASSIGN TO "GBSALES.DAT"
      ORGANIZATION IS LINE SEQUENTIAL.
SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".
```

```
DATA DIVISION.
FILE SECTION.
FD SalesFile.
01 SalesRecord.
   88 EndOfFile VALUE HIGH-VALUES.
   02 CityCode PIC 9.
   02 SalesPersonNum PIC 9.
   02 ValueOfSale PIC 9(4)V99.
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REPORT IS SalesReport.
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WORKING-STORAGE SECTION.
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   02 TableValues.
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```
REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

The FD entry of an ordinary print file would be followed by a record description of the print line such as -

```
FD PrintFile.
01 PrintLine PIC X(80).
```

```
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```
ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
```

```
SELECT SalesFile ASSIGN TO "GBSALES.DAT"
      ORGANIZATION IS LINE SEQUENTIAL.
SELECT PrintFile ASSIGN TO "SALESREPORT2.LPT".
```

```
DATA DIVISION.
FILE SECTION.
FD SalesFile.
01 SalesRecord.
   88 EndOfFile VALUE HIGH-VALUES.
   02 CityCode PIC 9.
   02 SalesPersonNum PIC 9.
   02 ValueOfSale PIC 9(4)V99.
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```
FD PrintFile
REPORT IS SalesReport.
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```
WORKING-STORAGE SECTION.
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```
01 NameTable.
   02 TableValues.
      03 FILLER PIC X(18) VALUE "Dublin Belfast ".
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```
REPORT SECTION.
```

```
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.
```

But in the Report Writer the
FD entry contains a clause
linking the file name with a
Report described in the
REPORT SECTION.

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

```

In this case it is a PAGE
HEADING group.

Note that we do not need to give
the group record a name if we
don't refer to it later.

01 TYPE IS PAGE HEADING.

```

02 LINE 1.
   03 COLUMN 12      PIC X(32)
                       VALUE "An e

```

```

02 LINE 2.
   03 COLUMN 6       PIC X(17)
                       VALUE "Bible Salesperson".
   03 COLUMN 23      PIC X(26)
                       VALUE " - Sales and Salary Report".

```

```

02 LINE 4.
   03 COLUMN 2       PIC X(4)  VALUE "City".
   03 COLUMN 12      PIC X(11) VALUE "Salesperson".
   03 COLUMN 28      PIC X(4)  VALUE "Sale".

```

```

02 LINE 5.
   03 COLUMN 2       PIC X(4)  VALUE "Name".
   03 COLUMN 13      PIC X(6)  VALUE "Number".
   03 COLUMN 28      PIC X(5)  VALUE "Value".

```

01 DetailLine TYPE IS DETAIL.

```

02 LINE IS PLUS 1.
   03 COLUMN 1       PIC X(9)
                       SOURCE CityName(CityCode).
   03 COLUMN 15      PIC 9
                       SOURCE SalesPersonNum.
   03 COLUMN 25      PIC $$,$$$$.99 SOURCE ValueOfSale.

```

01 SalesPersonGrp

TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

```

02 LINE IS PLUS 1.
   03 COLUMN 15      PIC X(21)
                       VALUE "Sales for salesperson".
   03 COLUMN 37      PIC 9 SOURCE SalesPersonNum.
   03 COLUMN 43      PIC X VALUE "=".
   03 SMS COLUMN 45  PIC $$$$,,$$$$.99 SUM ValueOfSale.

```

01 TYPE IS PAGE FOOTING.

```

02 LINE IS 53.
   03 COLUMN 1       PIC X(29)
                       VALUE "Programmer - Michael Coughlan".
   03 COLUMN 45      PIC X(6) VALUE "Page :".
   03 COLUMN 52      PIC Z9 SOURCE PAGE-COUNTER.

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

We want the page heading printed on the same line on each new page so we explicitly state the line number.

```

01 TYPE IS PAGE HEADING.
02 LINE 1.
   03 COLUMN 12      PIC X(32)
                        VALUE "An example"

02 LINE 2.
   03 COLUMN 6       PIC X(17)
                        VALUE "Bible Salesperson".
   03 COLUMN 23      PIC X(26)
                        VALUE " - Sales and Salary Report".

02 LINE 4.
   03 COLUMN 2       PIC X(4)  VALUE "City".
   03 COLUMN 12      PIC X(11) VALUE "Salesperson".
   03 COLUMN 28      PIC X(4)  VALUE "Sale".

02 LINE 5.
   03 COLUMN 2       PIC X(4)  VALUE "Name".
   03 COLUMN 13      PIC X(6)  VALUE "Number".
   03 COLUMN 28      PIC X(5)  VALUE "Value".

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
   03 COLUMN 1       PIC X(9)
                        SOURCE CityName(CityCode).
   03 COLUMN 15      PIC 9
                        SOURCE SalesPersonNum.
   03 COLUMN 25      PIC $$,$$$.$99 SOURCE ValueOfSale.

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
   03 COLUMN 15      PIC X(21)
                        VALUE "Sales for salesperson".
   03 COLUMN 37      PIC 9 SOURCE SalesPersonNum.
   03 COLUMN 43      PIC X VALUE "=".
   03 SMS COLUMN 45  PIC $$$,$$$.$99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
   03 COLUMN 1       PIC X(29)
                        VALUE "Programmer - Michael Coughlan".
   03 COLUMN 45      PIC X(6) VALUE "Page :".
   03 COLUMN 52      PIC Z9 SOURCE PAGE-COUNTER.

```

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
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FOOTING 52.

```

The COLUMN [IS] phrase allows us to position the characters in each line.

Scroll up to the report page in the previous section to see the result of these phrases.

```

01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
   VALUE "An example"

02 LINE 2.
03 COLUMN 6 PIC X(17)
   VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
   VALUE " - Sales and Salary Report".

02 LINE 4.
03 COLUMN 2 PIC X(4) VALUE "City".
03 COLUMN 12 PIC X(11) VALUE "Salesperson".
03 COLUMN 28 PIC X(4) VALUE "Sale".

02 LINE 5.
03 COLUMN 2 PIC X(4) VALUE "Name".
03 COLUMN 13 PIC X(6) VALUE "Number".
03 COLUMN 28 PIC X(5) VALUE "Value".

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
   SOURCE CityName(CityCode).
03 COLUMN 15 PIC 9
   SOURCE SalesPersonNum.
03 COLUMN 25 PIC $$,$$$.$99 SOURCE ValueOfSale.

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
   VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC $$$,$$$.$99 SUM ValueOfSale.

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
03 COLUMN 1 PIC X(29)
   VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

```

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

```

01 TYPE IS PAGE HEADING.

```

02 LINE 1.
   03 COLUMN 12      PIC X(32)
                       VALUE "An example COBOL Report Program".

```

```

02 LINE 2.
   03 COLUMN 6       PIC X(17)
                       VALUE "Bible Salesperson".
   03 COLUMN 23      PIC X(26)

```

```

   VALUE " - Sales and Salary Report".

```

```

02 LINE 4.
   03 COLUMN 2       PIC X(4)  VALUE
   03 COLUMN 12      PIC X(11) VALUE
   03 COLUMN 28      PIC X(4)  VALUE

```

```

02 LINE 5.
   03 COLUMN 2       PIC X(4)  VALUE
   03 COLUMN 13      PIC X(6)  VALUE
   03 COLUMN 28      PIC X(5)  VALUE

```

01 DetailLine TYPE IS DETAIL.

```

02 LINE IS PLUS 1.
   03 COLUMN 1       PIC X(9)
                       SOURCE City
   03 COLUMN 15      PIC 9
                       SOURCE SalesPersonNum.
   03 COLUMN 25      PIC $$,$$$.$99 SOURCE ValueOfSale.

```

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

```

02 LINE IS PLUS 1.
   03 COLUMN 15      PIC X(21)
                       VALUE "Sales for salesperson".
   03 COLUMN 37      PIC 9 SOURCE SalesPersonNum.
   03 COLUMN 43      PIC X VALUE "=".
   03 SMS COLUMN 45  PIC $$$,$$$.$99 SUM ValueOfSale.

```

01 **TYPE IS PAGE FOOTING.**

```

02 LINE IS 53.
   03 COLUMN 1       PIC X(29)
                       VALUE "Programmer - Michael Coughlan".
   03 COLUMN 45      PIC X(6) VALUE "Page :".
   03 COLUMN 52      PIC Z9 SOURCE PAGE-COUNTER.

```

Now let's specify the report group required to print the page footing.

As before we must specify the type of report group.
It is a PAGE FOOTING group.

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

```

```

01 TYPE IS PAGE HEADING.

```

```

02 LINE 1.
03 COLUMN 12      PIC X(32)
                   VALUE "An exa

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```

02 LINE 2.
03 COLUMN 6        PIC X(17)
                   VALUE "Bible Salesperson"
03 COLUMN 23       PIC X(26)
                   VALUE " - Sales and Salari

```

```

02 LINE 4.
03 COLUMN 2        PIC X(4)
03 COLUMN 12       PIC X(11)
03 COLUMN 28       PIC X(4)

```

```

02 LINE 5.
03 COLUMN 2        PIC X(4)
03 COLUMN 13       PIC X(6)
03 COLUMN 28       PIC X(5)

```

```

01 DetailLine TYPE IS DETAIL.

```

```

02 LINE IS PLUS 1.
03 COLUMN 1        PIC X(9)
                   SOURCE City
03 COLUMN 15       PIC 9
                   SOURCE Sale
03 COLUMN 25       PIC $$,$$$

```

```

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPe

```

```

02 LINE IS PLUS 1.
03 COLUMN 15       PIC X(21)
                   VALUE "Sale
03 COLUMN 37       PIC 9 SOURCE
03 COLUMN 43       PIC X VALUE
03 SMS COLUMN 45   PIC $$$$, $

```

```

01 TYPE IS PAGE FOOTING.

```

```

02 LINE IS 53.
03 COLUMN 1        PIC X(29)
                   VALUE "Programmer - Michael Coughlan".
03 COLUMN 45       PIC X(6) VALUE "Page :".
03 COLUMN 52       PIC Z9 SOURCE PAGE-COUNTER.

```

The **SOURCE** phrase allows a programmer to tell the Report Writer **where** to find the value to be printed.

We want to print a page number in the page footing and we do this by using the **SOURCE PAGE-COUNTER** phrase.

PAGE-COUNTER is a special register that allows programs to access the page count.

Each report in the Report Section has an associated **PAGE-COUNTER**

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
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01 TYPE IS PAGE HEADING.
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02 LINE 1.
   03 COLUMN 12      PIC X(32)
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                        VALUE " - Sales and Sal
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```
02 LINE 4.
   03 COLUMN 2        PIC X(4)
   03 COLUMN 12       PIC X(11)
   03 COLUMN 28       PIC X(4)
```

```
02 LINE 5.
   03 COLUMN 2        PIC X(4)
   03 COLUMN 13       PIC X(6)
   03 COLUMN 28       PIC X(5) VALUE "Value".
```

The LINE IS PLUS 1 phrase prints the sales information on the next line unless a new page is required.

If a new page is required the line is printed on the next page.

```
01 DetailLine TYPE IS DETAIL.
```

```
02 LINE IS PLUS 1.
   03 COLUMN 1        PIC X(9)
                        SOURCE CityName(CityCode).
   03 COLUMN 15       PIC 9
                        SOURCE SalesPersonNum.
   03 COLUMN 25       PIC $$,$$$.$99 SOURCE ValueOfSale.
```

```
01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
```

```
02 LINE IS PLUS 1.
   03 COLUMN 15       PIC X(21)
                        VALUE "Sales for salesperson".
   03 COLUMN 37       PIC 9 SOURCE SalesPersonNum.
   03 COLUMN 43       PIC X VALUE "=" .
   03 SMS COLUMN 45   PIC $$$,$$$.$99 SUM ValueOfSale.
```

```
01 TYPE IS PAGE FOOTING.
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```
02 LINE IS 53.
   03 COLUMN 1        PIC X(29)
                        VALUE "Programmer - Michael Coughlan".
   03 COLUMN 45       PIC X(6) VALUE "Page :".
   03 COLUMN 52       PIC Z9 SOURCE PAGE-COUNTER.
```

REPORT SECTION.
RD SalesReport
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03 COLUMN 12 PIC X(32)
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03 COLUMN 25 PIC \$\$,\$\$\$.\$99 SOURCE ValueOfSale.

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01 TYPE IS PAGE FOOTING.

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03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".

03 COLUMN 45 PIC X(6) VALUE "Page :".

03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

PAGE LIMIT specifies the size of the page.

HEADING specifies where the first heading line can be printed.

FOOTING specifies where the first footing line can be printed.

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
FIRST DETAIL 6
LAST DETAIL 42
FOOTING 52.

```

```
01 TYPE IS PAGE HEADING.
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02 LINE 1.
03 COLUMN 12      PIC X(32)
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```
02 LINE 4.
03 COLUMN 2        PIC X(4)
03 COLUMN 12       PIC X(11)
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```

```
02 LINE 5.
03 COLUMN 2        PIC X(4)
03 COLUMN 13       PIC X(6)
03 COLUMN 28       PIC X(5)
```

```
01 DetailLine TYPE IS DETAIL.
```

```
02 LINE IS PLUS 1.
03 COLUMN 1        PIC X(9)
                   SOURCE Ci
03 COLUMN 15       PIC 9
                   SOURCE Sa
03 COLUMN 25       PIC $$$,$$$.$99 SOURCE ValueOfSale.
```

When the salesperson number changes we want to print the total sales for the salesperson.

We can set up a report group (as shown) that describes **what** to print each time the salesperson number changes.

But how will the Report Writer know **when** to print this group?

01 SalesPersonGrp

```
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.
02 LINE IS PLUS 1.
03 COLUMN 15       PIC X(21)
                   VALUE "Sales for salesperson".
03 COLUMN 37       PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43       PIC X VALUE "=".
03 SMS COLUMN 45   PIC $$$,$$$.$99 SUM ValueOfSale.
```

```
01 TYPE IS PAGE FOOTING.
```

```
02 LINE IS 53.
03 COLUMN 1        PIC X(29)
                   VALUE "Programmer - Michael Coughlan".
03 COLUMN 45       PIC X(6) VALUE "Page :".
03 COLUMN 52       PIC Z9 SOURCE PAGE-COUNTER.
```

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
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01 TYPE IS PAGE HEADING.
02 LINE 1.
    03 COLUMN 12      PIC X(32)
                        VALUE "An example COBOL Report Program".

```

```

02 LINE 2.
    03 COLUMN 6       PIC X(17)
                        VALUE "Bible Salesperson".
    03 COLUMN 23      PIC X(26)
                        VALUE " - Sales and Sal

```

```

02 LINE 4.
    03 COLUMN 2       PIC X(4)
    03 COLUMN 12      PIC X(11)
    03 COLUMN 28      PIC X(4)

```

```

02 LINE 5.
    03 COLUMN 2       PIC X(4)
    03 COLUMN 13      PIC X(6)
    03 COLUMN 28      PIC X(5)

```

```

01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
    03 COLUMN 1       PIC X(9)
                        SOURCE Ci
    03 COLUMN 15      PIC 9
                        SOURCE Sa
    03 COLUMN 25      PIC $$,$$

```

```

01 SalesPersonGrp
TYPE IS CONTROL FOOTING Sales

```

```

02 LINE IS PLUS 1.
    03 COLUMN 15      PIC X(21)
                        VALUE "Sales for salesperson".
    03 COLUMN 37      PIC 9 SOURCE SalesPersonNum.
    03 COLUMN 43      PIC X VALUE "="
    03 SMS COLUMN 45  PIC $$$$,,$$.99 SUM ValueOfSale.

```

```

01 TYPE IS PAGE FOOTING.
02 LINE IS 53.
    03 COLUMN 1       PIC X(29)
                        VALUE "Programmer - Michael Coughlan".
    03 COLUMN 45      PIC X(6) VALUE "Page :".
    03 COLUMN 52      PIC Z9 SOURCE PAGE-COUNTER.

```

When we specify that a group is a control group, we must also specify the name of the control break data-item. The Report Writer monitors the value in this item and when it changes a control break occurs.

But if there is more than one control break item how does the Report Writer establish the relationship between them?

```

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
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01 TYPE IS PAGE HEADING.
02 LINE 1.
03 COLUMN 12 PIC X(32)
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```

02 LINE 2.
03 COLUMN 6 PIC X(17)
VALUE "Bible Salesperson".
03 COLUMN 23 PIC X(26)
VALUE " - Sales and Salary Report"

```

```

02 LINE 4.
03 COLUMN 2 PIC X(4)
03 COLUMN 12 PIC X(11)
03 COLUMN 28 PIC X(4)

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02 LINE 5.
03 COLUMN 2 PIC X(4)
03 COLUMN 13 PIC X(6)
03 COLUMN 28 PIC X(5)

```

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01 DetailLine TYPE IS DETAIL.
02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE Ci
03 COLUMN 15 PIC 9
SOURCE Sa
03 COLUMN 25 PIC $$$,$$

```

```

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

```

```

02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
VALUE "Sales for salesperson".
03 COLUMN 37 PIC 9 SOURCE SalesPersonNum.
03 COLUMN 43 PIC X VALUE "=".
03 SMS COLUMN 45 PIC $$$,$$,99 SUM ValueOfSale.

```

```

01 TYPE IS PAGE FOOTING.

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02 LINE IS 53.
03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

```

The order of the control breaks is established by the **CONTROLS ARE** phrase in the report's RD entry. The control break items are given in order of seniority. If a senior control break occurs, it automatically causes a control break on all items junior to it.

REPORT SECTION.
RD SalesReport
CONTROLS ARE SalesPersonNum
PAGE LIMIT IS 66
HEADING 1
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01 TYPE IS PAGE HEADING.

02 LINE 1.
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02 LINE 5.
03 COLUMN 2 PIC X(4)
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01 DetailLine TYPE IS DETAIL.

02 LINE IS PLUS 1.
03 COLUMN 1 PIC X(9)
SOURCE Ci
03 COLUMN 15 PIC 9
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03 COLUMN 25 PIC \$\$\$,\$\$

01 SalesPersonGrp
TYPE IS CONTROL FOOTING SalesPersonNum NEXT GROUP PLUS 2.

02 LINE IS PLUS 1.
03 COLUMN 15 PIC X(21)
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01 TYPE IS PAGE FOOTING.

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03 COLUMN 1 PIC X(29)
VALUE "Programmer - Michael Coughlan".
03 COLUMN 45 PIC X(6) VALUE "Page :".
03 COLUMN 52 PIC Z9 SOURCE PAGE-COUNTER.

We have identified the item that contains the salesperson total with the name SMS.

In this program we don't actually need to identify this item because we never refer to it but it is declared now because we will refer to it when we extend the program.

```
PROCEDURE DIVISION.  
Begin.  
    OPEN INPUT SalesFile.  
    OPEN OUTPUT PrintFile.  
    READ SalesFile  
        AT END SET EndOfFile TO TRUE  
    END-READ.  
INITIATE SalesReport.  
    PERFORM PrintSalaryReport  
        UNTIL EndOfFile.  
TERMINATE SalesReport.  
    CLOSE SalesFile, PrintFile.  
    STOP RUN.
```

```
PrintSalaryReport.  
GENERATE DetailLine.  
    READ SalesFile  
        AT END SET EndOfFile TO TRUE  
    END-READ.
```

The INITIATE statement sets up the report and in it we specify which report we want to set up.

The TERMINATE closes down the report and we must specify which report to close.

The GENERATE statement prints the detail lines and must specify the name we have assigned to the DETAIL group.

Each time we read a record we generate a detail line.

End of commentary